

A PARTIAL ANSWER TO PROBLEM G OF AVILES–PLEBANEK–RODRIGUEZ

AGENT_LANE_10

SOURCE PROBLEM

In arXiv:1208.2207, Aviles–Plebanek–Rodriguez ask in Problem G whether

$$\text{Ba}(C_p(K)) \neq \text{Ba}(C_w(K))$$

whenever K is infinite and $C(K)$ is a Grothendieck space. Here $C_p(K)$ and $C_w(K)$ denote $C(K)$ with the pointwise and weak topologies, respectively.

PARTIAL THEOREM

Theorem 1. *Let K be an infinite compact Hausdorff space. Suppose that $C(K)$ is a Grothendieck space and that every closed separable subspace of K is scattered. Then*

$$\text{Ba}(C_p(K)) \neq \text{Ba}(C_w(K)).$$

In particular, the conclusion holds if every closed separable subspace of K is countable.

PROOF INTUITION

The source paper proves that if the two Baire sigma-algebras are equal, then every probability measure on K must be concentrated on a closed separable subset of K . It also proves that, for infinite Grothendieck $C(K)$, there is a probability measure not concentrated on any countable subset of K . The added scatteredness hypothesis turns any closed separable carrier into a countably supported carrier, because Radon probabilities on scattered compacta are countably supported. These two facts are incompatible.

PROOF

Assume for contradiction that

$$\text{Ba}(C_p(K)) = \text{Ba}(C_w(K)).$$

By the criterion quoted at the start of Section 4 of arXiv:1208.2207 (from Rodriguez–Vera, Proposition 3.6), every $\text{Ba}(C_p(K))$ -measurable probability measure on K is concentrated on a closed separable subset of K . Since $\text{Ba}(C_w(K))$ is generated by the probability measures $P(K)$, the assumed equality implies that every $\mu \in P(K)$ is concentrated on some closed separable $F_\mu \subset K$.

By hypothesis F_μ is scattered. A standard theorem on measures on scattered compact spaces, used in the source paper via Semadeni [2, Theorem 19.7.6], says equivalently that every Radon probability measure on a scattered compact space is concentrated on a countable subset. Hence every $\mu \in P(K)$ is concentrated on a countable subset of K .

This contradicts Proposition 5.1 of arXiv:1208.2207. That proposition states that if K is infinite and $C(K)$ is Grothendieck, then $\text{Seq}(\text{co}\Delta_K) \neq P(K)$; its proof gives the stronger fact needed here: there exists a probability measure on K that is not concentrated on any countable subset of K . Therefore the Baire sigma-algebras cannot be equal.

Finally, if every closed separable subspace of K is countable, then it is scattered, so the previous conclusion applies.

VERIFICATION AND LIMITATIONS

The proof uses only statements explicitly present in the source paper plus the standard scattered-compact measure characterization cited there. This is a proper subcase of Problem G: it does not settle Grothendieck compacta that contain non-scattered closed separable subspaces.

REFERENCES

- [1] A. Aviles, G. Plebanek, and J. Rodriguez, *On Baire measurability in spaces of continuous functions*, arXiv:1208.2207.
- [2] Z. Semadeni, *Banach spaces of continuous functions. Vol. I*, PWN–Polish Scientific Publishers, Warsaw, 1971.